EE3621: Digital Signal Processing

Lecture 23: Digital Oscillators, Waveform Generation & NCO
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Need for Digital Oscillators in DSP (SDR, modems).
- **05:00 – 15:00 (10 mins):** Recursive Oscillator (IIR approach), derivations, and coupled equations.
- **15:00 – 20:00 (5 mins):** Poles on the unit circle, stability issues, amplitude drift.
- **20:00 – 25:00 (5 mins):** Amplitude correction methods (Rotor method).
- **25:00 – 32:00 (7 mins):** Numerically Controlled Oscillator (NCO) and DDS. Phase noise and spurs.
- **32:00 – 35:00 (3 mins):** Chirp signal generation (LFM).
- **35:00 – 40:00 (5 mins):** Checkpoint Questions and Summary.

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2 Need for Digital Oscillators

In modern digital systems, generating precise waveforms entirely in the digital domain is crucial. Digital oscillators replace bulky and temperature-sensitive analog Voltage-Controlled Oscillators (VCOs) with digital equivalents.

Applications include Software-Defined Radio (SDR) for digital up/down-conversion, Modems for carrier generation, and arbitrary waveform generation for test equipment. They offer exact reproducibility without thermal drift.

3 Recursive Oscillator (IIR Approach)

We can generate a sinusoid by realizing a discrete-time system whose impulse response is a sinusoid. Consider the complex exponential sequence:

$$y[n] = e^{j\omega_0 n} \tag{1}$$

Derivation of the difference equation:

1. Sum of exponentials at adjacent times $n+1$ and $n-1$:
$$y[n+1] + y[n-1] = e^{j\omega_0(n+1)} + e^{j\omega_0(n-1)}$$
2. Factor out $e^{j\omega_0 n}$:
$$y[n+1] + y[n-1] = e^{j\omega_0 n}(e^{j\omega_0} + e^{-j\omega_0})$$

3. Apply Euler's identity $e^{j\omega_0} + e^{-j\omega_0} = 2\cos(\omega_0)$:

$$y[n+1] + y[n-1] = 2\cos(\omega_0)y[n]$$

4. Shift time index by -1:

$$y[n] = 2\cos(\omega_0)y[n-1] - y[n-2] \tag{2}$$

To generate both sine and cosine without direct-form stability issues, we separate real and imaginary parts for coupled oscillator equations:

$$c[n] = 2\cos(\omega_0)c[n-1] - c[n-2] \tag{3}$$
$$s[n] = 2\cos(\omega_0)s[n-1] - s[n-2] \tag{4}$$

This only needs a multiplier and two unit delays per channel.

4 Poles on the Unit Circle and Amplitude Drift

Taking the Z-transform of the oscillator difference equation:

$$H(z) = \frac{z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}} \tag{5}$$

The roots of the denominator are strictly on the unit circle: $z = e^{\pm j\omega_0}$. Because the poles are precisely on the unit circle, the system is marginally stable. In fixed-point arithmetic, coefficient quantization and round-off noise cause the amplitude to drift exponentially (decay or grow).

5 Amplitude Correction

To combat drift, periodic normalization can be used, scaling states by the inverse of instantaneous amplitude. Alternatively, the Rotor method implements complex multiplication directly:

$$y[n] = e^{j\omega_0}y[n-1] \tag{6}$$

requiring only 4 real multiplies and 2 adds per sample.

6 Numerically Controlled Oscillator (NCO) & DDS

An NCO explicitly tracks phase and maps it to amplitude via a Lookup Table (LUT). At each clock f_s, we add the phase increment $\Delta\phi$:

$$\Phi = (\Phi + \Delta\phi) \pmod{2^B} \tag{7}$$

$$f_{out} = \frac{\Delta\phi \cdot f_s}{2^B} \tag{8}$$

Direct Digital Synthesis (DDS) combines an NCO with a DAC and LPF. **Phase Truncation & Spurs:** Because LUT addresses are truncated, predictable spurious tones appear at $f_{spur} = k \cdot f_s/2^B$. Dithering (adding random noise before truncation) is used to randomize these spurs and improve Spurious-Free Dynamic Range (SFDR).

7 Chirp Signal Generation

Linear frequency modulation (LFM):

$$x[n] = \cos\left(\omega_0 n + \frac{\mu}{2} n^2\right) \tag{9}$$

Generated by feeding an NCO with a linearly increasing tuning word. Applications include RADAR and sonar.

8 Illustrative Plots

For signal envelope bounding and spur suppression visualization, window functions are often applied:

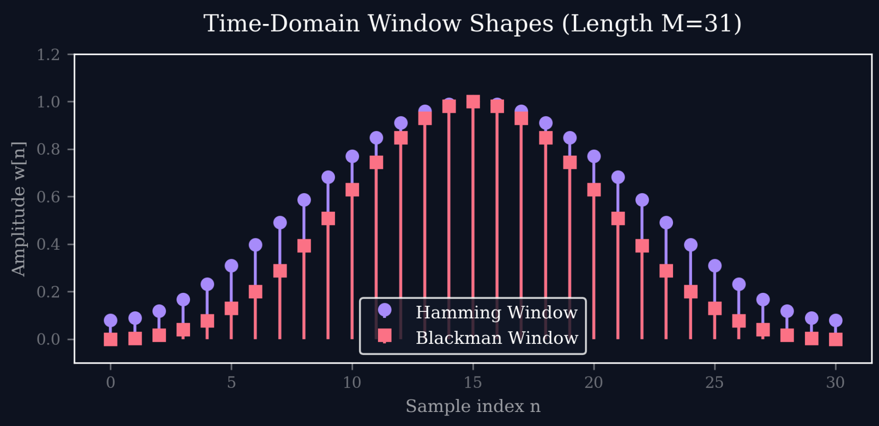

Figure 1: Oscillator output windowing functions.

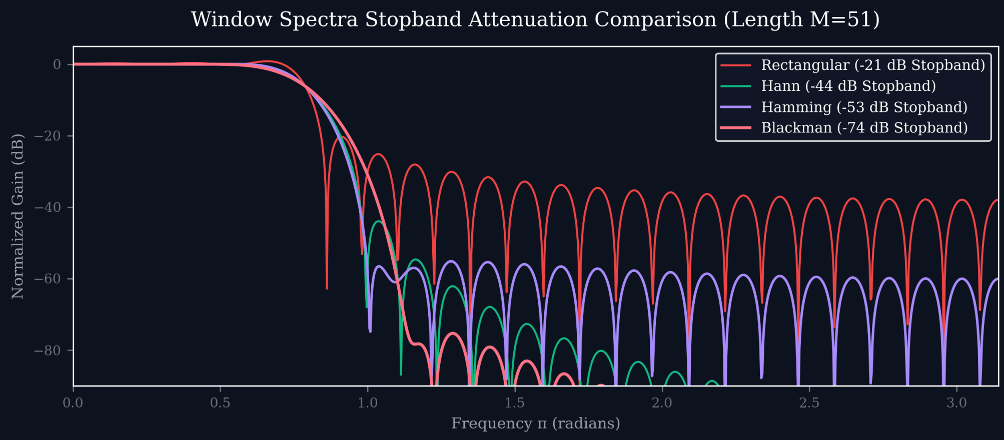

Figure 2: Spur suppression levels under different windows.

9 Checkpoints

Q1: Why does a recursive digital oscillator suffer from amplitude drift?
Answer: Poles must be precisely on the unit circle. Fixed-point coefficient quantization moves them off the circle, causing exponential growth or decay. Accumulated roundoff noise also contributes.

Q2: NCO with 32-bit accumulator, 100 MHz clock. Tuning word for 1 MHz?
Answer: $f_{out} = (\Delta\phi \cdot f_s)/2^B \implies \Delta\phi = (10^6 \cdot 2^{32})/10^8 \approx 42949673.$

Q3: Why dither an NCO phase?
Answer: It randomizes periodic phase truncation errors, spreading spur energy into the noise floor and improving SFDR.

10 Key Formulas

Concept	**Formula**
Recursive Oscillator	$y[n] = 2\cos(\omega_0)y[n-1] - y[n-2]$
Coupled Form	$c[n] = 2\cos(\omega_0)c[n-1] - c[n-2]$
Transfer Function	$H(z) = \frac{z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$
NCO Output Freq	$f_{out} = \frac{\Delta\phi \cdot f_s}{2^B}$
Chirp Phase	$\phi[n] = \omega_0 n + \frac{\mu}{2}n^2$